

BS 192

Undergraduate Science Laboratory (Physics)

Lab 5

Experiment 6. Ultrasonic Diffraction Experiment

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Introduction

The experiment of ultrasonic diffraction is used to find the bulk modulus and compressibility of a liquid by measuring the velocity of ultrasonic waves in the liquid. This is in turn found by finding the distance between the fringes in the diffraction pattern. This is a very interesting experiment which uses the concepts of diffraction (optics) in finding the properties of fluids (compressibility and bulk modulus).

Aim

The aim of this experiment is to:

- Determine the speed of the ultrasonic wave in liquid using ultrasonic diffraction apparatus
- Measure the bulk modulus of the given liquid
- Estimate the compressibility of the liquid

Apparatus Required

1. Kinematic Laser Mount
2. Diode Laser
3. Power Supply for Laser
4. Glass Tank with Liquid
5. Glass Tank Holder
6. Crystal with Mount
7. RF Oscillator
8. Optical Rail (1500 mm)
9. Cell mount with linear translation stage and Pinhole detector
10. Output measurement unit



Figure 1: Apparatus required for the experiment. Source [1]

Principles and Theory

The RF oscillator produces ultrasonic waves in the glass tank with liquid that gets reflected by the bottom of the tank, forming standing waves due to interference. The speed of these waves is given as $V = \vartheta\Lambda$ where ϑ is the frequency of the oscillator and Λ is the wavelength of these waves. Next, a laser is passed through the tank parallel to the surface of the plane face of the oscillator. The nodes of the standing wave act like the slits analogous to those in Young's Double Slit Experiment, causing an interference pattern on the detector stage. The detector can be moved on the stage with the help of a micrometer, enabling us to measure the change in intensity of the interference pattern at different distances D from the centre of the cell mount. This can be used to calculate the angle of diffraction for each observation $\theta = \tan^{-1}(D/L) \approx D/L$. A bright spot is observed at distance D_n when the rays passing through the nodes of the standing wave in liquid, have path difference $\Lambda \sin(\theta_n)$, namely as an integral (n) multiple of λ , i.e

$$\Lambda \sin \theta_n = n\lambda$$
$$\boxed{\Lambda = n\lambda / \sin \theta_n} \approx n\lambda L / D_n$$

With this, we can find the average value of $n\lambda / \sin \theta_n$ as a good approximation for the true value of Λ (among other things, like the statistical error). With the value of Λ and ϑ , we can calculate $V = \vartheta\Lambda$ and then the bulk modulus and adiabatic compressibility as:

$$B = \rho V^2$$
$$K = 1/B = \frac{1}{\rho V^2}$$

Experimental Setup and Procedure

Setup and Details of Apparatus

This apparatus is well designed for the ultrasonic diffraction experiment. We will refer to the equipment in figure 1, which was used by us. It has a glass container which holds the liquid whose parameters are to be measured. We also have an RF Oscillator dipped inside the container which vibrates according to the set frequency. These set vibrations induce the occurrence of standing waves in the container. Upon incidenting the container with laser, the nodes and antinodes are responsible for the creation of the fringe pattern on the detector device. The detector device is capable of measuring the intensity of light as we would use it to find the peaks in the fringe intensities. All the equipment is set along the optical axis as well as have micrometers at sufficient places to measure the distances required to be found.

Procedure

1. Fix the laser mount on the optical rail and place laser on it.
2. Place the glass tank holder on the rail and a filled tank with liquid on the tank holder.
3. Fix the piezo-electric crystal on the mount, immersed fully in liquid and connect it to the RF oscillator.
4. Fix cell mount with the linear translation stage on the rail.
5. Insert the pin-hole detector into the cell mount and connect probe to measurement unit.
6. Switch on the laser and make it parallel to the face of the piezo-electric crystal, and it is going through the wave in the liquid. Adjust and align the laser, tank and center of detector stage.
7. Adjust the frequency of the oscillator until a good fringe pattern is observed on both sides on the central bright spot on the detector stage.
8. Move the micrometer driven stage driven detector to the extreme end of the fringe pattern.
9. Using the micrometer, move the detector to different distances (starting from outermost bright ring to the central bright spot) from the center and take readings of

- The output of the detector (I)
- The distance from the central bright spot, that the micrometer is showing (D).

Results and Calculations

Calculations

- Plot I vs D graph
- Identify the points of local maxima in I(D), i.e. D_n corresponding to the n^{th} bright spot (ring) for all n possible using the data.
- Calculate $\theta_n = \tan^{-1}(D_n/L)$ for each n
- Calculate $\Lambda_n = n\lambda/\sin(\theta_n)$ for each n
- Calculate $V = \vartheta(\Lambda_n)$ for all n
- Calculate Bulk Modulus and Compressibility using:

$$B = \rho V^2$$

$$K = 1/B = \frac{1}{\rho V^2}$$

Raw Data

Micrometer Reading, D (mm)	Detector Output, I (μA)
1	0.2
1.25	0.1
1.5	0.2
1.75	0.2
2	0.2
2.25	0.2
2.5	0.3
2.75	0.3
3	0.4
3.25	0.6
3.5	0.3
3.75	0.3
4	0.4
4.25	0.5
4.5	0.7
4.75	0.6
5	0.7
5.25	0.5
5.5	0.6
5.75	0.6
6	0.6
6.25	0.6
6.5	0.7
6.75	0.6
7	1.1
7.25	0.8
7.5	1.2
7.75	1.4

8	1.5
8.25	1.2
8.5	1.3
8.75	1.4
9	2.2
9.25	1.8
9.5	2.4
9.75	3.5
10	6.3
10.25	5
10.5	5.9
10.75	4.8
11	16
11.25	35.7
11.5	74.8
11.75	33
12	35.1
12.25	61.2
12.5	141.8
12.75	4.00E+02
13	1.30E+03
13.25	1.00E+03
13.5	128
13.75	3.00E+02
14	9.00E+02
14.25	3.00E+03
14.5	7.80E+03
14.75	9.50E+03
15	1.80E+03
15.25	3.00E+02
15.5	115
15.75	3.00E+02
16	7.00E+02
16.25	1.10E+03
16.5	4.00E+02
16.75	53.1
17	31.8
17.25	26.1
17.5	31.6
17.75	65.6
18	47.3
18.25	8.7
18.5	6
18.75	4.1
19	3.5
19.25	4.1

19.5	3.6
19.75	2.2
20	1.3
20.25	1
20.5	0.7
20.75	0.5
21	0.5
21.25	0.3
21.5	0.2
21.75	0.3
22	0.3
22.25	0.5
22.5	0.2
22.75	0.2
23	0.1
23.25	0.1
23.5	0.1
23.75	0.1
24	0.1

Data Processing and Tabulation

A total of about 100 data points were collected throughout the experiment.

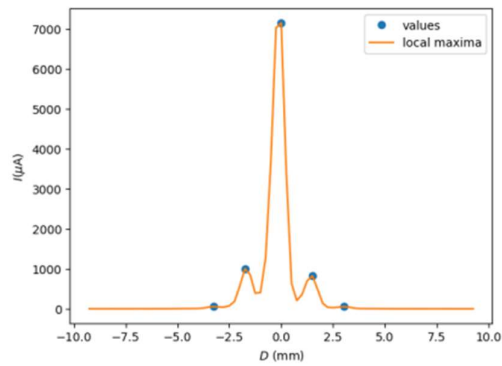


Figure 2: Plot of I against D

The values of the peaks thus obtained are:

N	D_n (mm)
-2	-3.25
-1	-1.75
0	0
1	1.5
2	3

Note that the sign indicates whether the micrometer reading was less than the reading for central fringe, or more, nothing more.

The set value of Frequency $\vartheta = 3.05\text{MHz}$

$L = 1370 - 276 \text{ mm} = 1094 \text{ mm}$

zero error MSR = 14.5

zero error CSR = 8

$\Lambda = 650 \text{ nm}$

Least Count of Micrometer = 0.01 mm

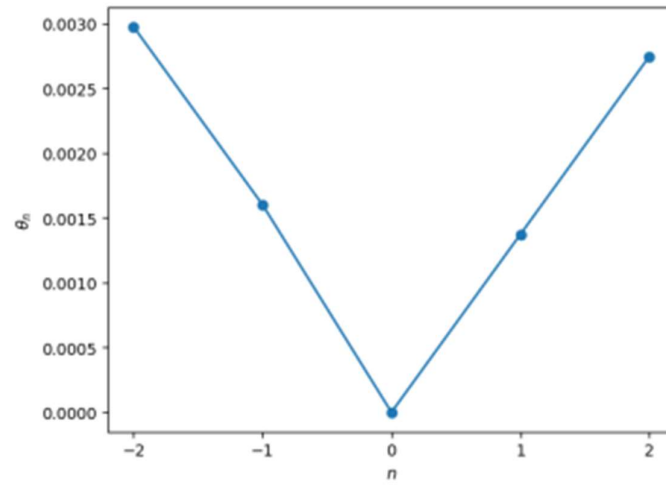


Figure 3: Plot of θ_n in radians for different values of n

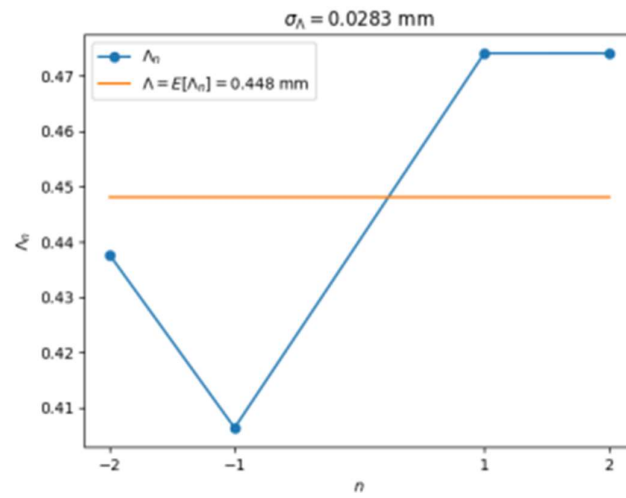


Figure 4: Plot of Λ_n (mm) against n

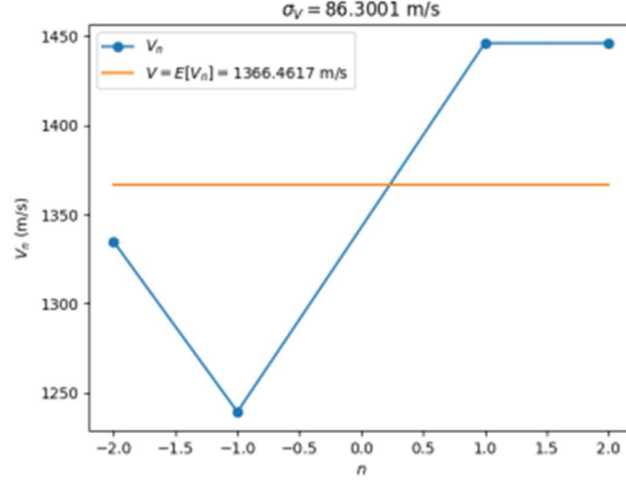


Figure 5: Plot of V_n against n

So, the expected value of V is 1366.46 ± 86.30 m/s.

Final Results and Error Analysis

Error 1: Least Count Error

Since we have $\Lambda_n \approx n\lambda L/D_n$, so we can say $\Delta_{LC}\Lambda_n/\Lambda_n \approx \Delta_{LC}D_n/D_n = 0.5LC/D_n$. This gives us $\Delta_{LC}\Lambda_n \approx \frac{1}{2}n\lambda LD_n^{-2} \times LC$, which enables use to calculate $\Delta_{LC}\Lambda = \Delta_{LC}\Lambda_n$. This value turns out to be around 0.001 mm, which is negligible compared to the standard deviation. This in turn gives a very small $\Delta_{LC}V$ compared to σ_V , and can thus be discarded, along with the least count errors in β and K .

Error 2: Standard Deviation Error

The statistical standard deviation error is 86.30 m/s in the velocity of sound in the given medium. This leads to a statistical error of 235.85×10^6 Pa in the Bulk Modulus and $6.76 \times 10^{-11} \text{Pa}^{-1}$ in compressibility of the given medium.

Final Value with Errors from our Experiment

The final value of the velocity of ultrasonic waves in the medium was 1366.46 ± 86.30 m/s.

The final value of the Bulk Modulus of the medium was $1.87 \times 10^9 \pm 235.85 \times 10^6$ Pa.

The final value of the Compressibility of the medium was $5.36 \times 10^{-10} \pm 6.67 \times 10^{-11} \text{Pa}^{-1}$.

Discussion and Conclusion

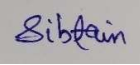
Assuming that the liquid was water, which has a speed of sound to be 1481 m/s [2], bulk modulus of 2.1 GPa [3], and thus a compressibility of $4.74 \times 10^{-10} \text{Pa}^{-1}$, our calculated values are slightly outside of the error margin. One probable reason for this deviation might be the slight inaccuracies in the measurements or the dust that might have dissolved in the water, or some error in the set-up. Another hypothesis might be that the liquid wasn't pure water but had something dissolved in it.

Author Contributions

Jaidev	Experimentation, Data Collection, Report Development
Pranav	Data Processing and Analysis, Report Development

Sibtain	Experimentation and Data Collection, Report Review
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Digitally Signed by:

References

[1] Lab Manual

[2] https://en.wikipedia.org/wiki/Speed_of_sound

[3] https://en.wikipedia.org/wiki/Bulk_modulus